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The Tactical Substrate

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Abstract

The C5ISR edge is not an extension of the enterprise network. It is a fundamentally different operating environment where the WAN link is intermittent by design, blackout windows are not anomalies but the expected envelope, and session-oriented protocols fail structurally rather than occasionally. This paper provides the survey frame for *The Implications of Edge Degraded Ops*. It names the substrate: Delay-Tolerant Networking bundle custody, write-ahead-log replication, CRDT-based state coherence, and an out-of-path AI Supervisor operating under the HGC³AE² human-governance constraint. Nine problem-unpack papers follow this frame; the governor-lens paper provides the HGC³AE² treatment. A senior architect who reads only this paper can defend the architecture at a technical briefing. Every other paper in the series is the detailed argument for one of the claims made here.

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ABSTRACT

Mission-ready agentic systems will, with growing frequency, operate outside the connectivity, capacity, and governance envelopes that hyperscaler-default deployment presumes. The **tactical substrate** is the architectural commitment that names what those out-of-envelope conditions are, why they break the operating assumptions of cloud-native AI, and what an agentic system must satisfy to remain mission-ready under them. This paper is the frame paper for the Edge Degraded Operations Undecalogy: it specifies the substrate over which the remaining ten papers operate. The substrate is characterized by **disconnected, intermittent, and limited (DDIL)** connectivity profiles; by **degraded compute** that may oscillate between substantial and severely constrained; by **partial governance reach** when authoritative governance state cannot be continuously consulted; and by **adversarial pressure** that hyperscaler-default deployments assume away. The Undecalogy's argument is that agentic systems can be architected to operate in this substrate without sacrificing the structural commitments of HGC³AE²; the present paper specifies the substrate that argument operates over.¹²

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1. INTRODUCTION

Cloud-native AI presupposes a substrate: stable broadband connectivity, elastic compute capacity, accessible centralized governance, and an operating environment shaped by commercial rather than adversarial pressures.³ The substrate is real and operationally valuable; for the workload mix that has historically dominated cloud computing, the substrate's properties match the workload's requirements. The Mission-Ready AI Decalogy's architecture, including its federation pattern in MR-P9, operates well on this substrate.

The Edge Degraded Operations substrate is structurally different. Three classes of operational environment fall outside the cloud-native presumption. **Tactical** environments — military, public-safety, expeditionary humanitarian — operate under degraded connectivity, with adversarial pressure on every layer of the stack, and with authoritative governance that cannot be assumed continuously available.⁴⁵ **Industrial** environments — process control, transportation, energy systems — operate with safety-critical determinism requirements and

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 Weisong Shi et al., "Edge Computing: Vision and Challenges," *IEEE Internet of Things Journal* 3, no. 5 (2016): 637–646, <https://doi.org/10.1109/JIOT.2016.2579198>.

3 Michael Armbrust et al., "A View of Cloud Computing," *Communications of the ACM* 53, no. 4 (2010): 50–58, <https://doi.org/10.1145/1721654.1721672>.

4 Michael C. Atighetchi et al., "Survivable Information Storage Systems," *IEEE Computer* 33, no. 8 (2000): 61–68, <https://doi.org/10.1109/2.863969>.

5 Suraj Yatnalli et al., "Tactical Edge Computing: A Survey," *IEEE Internet of Things Journal* 9, no. 13 (2022): 10453–10472, <https://doi.org/10.1109/JIOT.2022.3158876>.



connectivity profiles that hyperscaler SLAs do not address. **Constrained** environments — IoT, embedded systems, deep-edge sensor networks — operate at orders of magnitude less capacity than cloud workloads presume.⁶

These three classes share structure: each requires the agentic operating model to function on substrate properties that the cloud-native architectural presumption does not satisfy. The argument of the Edge Degraded Operations Undecalogy is that the agentic operating model can be architected to operate in this substrate while preserving HGC³AE²'s structural commitments. The present paper specifies the substrate that the remaining ten papers in the series develop.

The paper proceeds in canonical Mission-Ready shape. §2 names five operational failure modes that cloud-native deployment exhibits when deployed onto the tactical substrate. §3 specifies the substrate's four-property characterization. §4 treats substrate operation as a continuous process under DDIL conditions. §5 binds the substrate to Skipjack governance. §6 returns to HGC³AE² and frames the Undecalogy's remaining papers.

2. FAILURE MODES OF CLOUD-NATIVE AI ON THE TACTICAL SUBSTRATE

Five operational failure modes recur when cloud-native architectures are deployed onto the tactical substrate without architectural adaptation.

Connectivity-dependent inference loops. Cloud-native AI typically routes inference to centralized endpoints. When connectivity degrades, inference fails. The system halts; the operator has no agentic capability until connectivity returns. The DDIL operational requirement is that inference continue under connectivity degradation, not fail with it.⁷

Governance-state staleness. Cloud-native governance presumes authoritative state is continuously consultable. The tactical substrate periodically severs this consultation. The agentic system must either operate against stale governance state (risky) or halt until governance is reachable (also risky if mission criticality is high). The architectural response, developed across the Undecalogy, is pre-authorized envelopes that remain valid under disconnection.⁸

Persistence-tier divergence. When connectivity is intermittent, multiple substrate instances may evolve persistent state independently. Reconciliation on reconnection is non-trivial. The cloud-native presumption of a single authoritative tier does not hold; the substrate must coordinate state across boundaries that may be active or quiescent at any moment.⁹

Capacity oscillation. Cloud-native deployment assumes capacity is elastically available. The tactical substrate experiences capacity that oscillates between adequate, marginal, and

6 Yansong Shi et al., "Edge AI: On-Demand Accelerating Deep Neural Network Inference via Edge Computing," *IEEE Transactions on Wireless Communications* 19, no. 1 (2020): 447–457, <https://doi.org/10.1109/TWC.2019.2946140>.

7 Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007, <https://datatracker.ietf.org/doc/rfc4838/>.

8 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

9 Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski, "Conflict-Free Replicated Data Types," *Symposium on Self-Stabilizing Systems (SSS '11)*, 2011, https://doi.org/10.1007/978-3-642-24550-3_29.



severely constrained on operational time scales. Workloads must degrade gracefully under capacity loss, not fail catastrophically.

Adversarial pressure on every layer. Cloud-native deployment assumes the substrate is commercially adversarial (other tenants, occasional malicious actors) but operationally benign. The tactical substrate assumes the substrate is operationally adversarial — substrate compromise, communication interception, governance impersonation are first-class concerns at every layer.¹⁰

The unified problem: **the tactical substrate's properties differ from the cloud-native substrate in ways that produce structural failure modes, not edge cases.**

3. THE FOUR-PROPERTY SUBSTRATE CHARACTERIZATION

The tactical substrate is characterized by four properties that operate continuously: **DDIL connectivity, degraded compute, partial governance reach, and adversarial pressure.**

DDIL connectivity. Disconnected, intermittent, and limited connectivity is the substrate's default rather than an exception. The agentic system must operate with the expectation that any single point of connectivity will fail, that intermittent reconnections will be the principal communication mode, and that limited bandwidth will constrain what can be transmitted when connectivity is available.¹¹¹²

Degraded compute. Compute capacity oscillates. Tactical edge nodes may have substantial accelerator capacity (forward-deployed Omen-class hardware) but operate under power, thermal, and bandwidth constraints that the same hardware would not face in a datacenter. The agentic system must operate within actual current capacity, not against an assumed peak.

Partial governance reach. Authoritative governance — the human-led envelope updates that HGC³AE² requires — cannot be continuously consulted. The substrate must therefore satisfy the H of HGC³AE² through *pre-authorized envelopes* that survive disconnection from the authoritative governance source, with explicit constraints on what the system may do while disconnected and on what it must do upon reconnection.¹³

Adversarial pressure. Every layer of the substrate is under adversarial pressure: cryptographic, electromagnetic, supply-chain, operational. The architecture cannot assume away these pressures; it must architect for them as first-class operating conditions.¹⁴¹⁵

10 Jeffrey M. Voas and Phillip A. Laplante, "The IoT Blame Game," *IEEE Computer* 50, no. 6 (2017): 69–73, <https://doi.org/10.1109/MC.2017.169>.

11 Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007, <https://datatracker.ietf.org/doc/rfc4838/>.

12 Forrest Warthman, "Delay- and Disruption-Tolerant Networks (DTNs): A Tutorial," InterPlanetary Networking Special Interest Group (IPNSIG); supplemented by Kevin Fall, "A Delay-Tolerant Network Architecture for Challenged Internets," *Proceedings of ACM SIGCOMM '03*, 2003, <https://doi.org/10.1145/863955.863960>.

13 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

14 Jeffrey M. Voas and Phillip A. Laplante, "The IoT Blame Game," *IEEE Computer* 50, no. 6 (2017): 69–73, <https://doi.org/10.1109/MC.2017.169>.



The four properties interact. DDIL connectivity multiplies the consequences of partial governance reach. Degraded compute interacts with adversarial pressure to produce capacity-exhaustion attacks. The architecture's response is in the remaining ten papers of the Undecalogy.

4. SUBSTRATE OPERATION AS A CONTINUOUS PROCESS

Operation on the tactical substrate is continuous through three recurring phases.

Disconnected operation. The substrate operates against pre-authorized envelopes when authoritative governance is not reachable. The cognitive node (MR-P4) is the unit of disconnected operation; its five pillars are what enable continued operation without continuous governance consultation.¹⁶

Reconciliation. Upon reconnection, the substrate reconciles its accumulated state with the authoritative governance source. Reconciliation includes envelope updates, persistent-state merging, and surfacing of governance events that occurred during disconnection. The CRDT and WAL primitives developed in EDO-P4 are the operational instruments.¹⁷

Adjustment. The governance layer adjusts envelopes based on what reconciliation revealed: drift, anomaly, expected-versus-observed behavior. Adjusted envelopes propagate to the substrate for the next operational cycle.

The cycle is the HGC³AE² exponent operating under DDIL conditions: telemetry feeds adjustment feeds envelope refinement feeds the next operational cycle.

5. SKIPJACK COUPLING UNDER DDIL CONDITIONS

The Skipjack Protocol's four enforcement mechanisms operate under DDIL conditions with adaptations.¹⁸ **Context integrity** holds within the disconnected substrate (the substrate enforces integrity of its own state) and across reconnection (reconciliation verifies and merges). **Structured routing** operates within the substrate's available capacity and within the pre-authorized envelope. **HITL control points** fire when the substrate detects approach to envelope boundaries; under disconnection, the control point becomes a *halt-and-await-reconnection* event rather than a *halt-and-await-human* event. **Iterative feedback** closes asynchronously across the reconnection cycle.

The mechanisms are not weaker under DDIL; they are adapted. The tactical substrate's architecture preserves Skipjack's commitments through pre-authorization, integrity discipline, and reconciliation rather than through continuous availability of authoritative governance.

15 Ali Humayed, Jingqiang Lin, Fengjun Li, and Bo Luo, "Cyber-Physical Systems Security — A Survey," *IEEE Internet of Things Journal* 4, no. 6 (2017): 1802–1831, <https://doi.org/10.1109/JIOT.2017.2703172>.

16 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

17 EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

18 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



6. IMPLICATIONS AND CONCLUSION

The implications of the tactical substrate's properties extend to architectural investment (pre-authorized envelopes, CRDT persistence, reconciliation tooling are each first-class concerns), operational doctrine (DDIL operations is a discipline distinct from cloud-native operations), and the broader question of where governed agentic operation can be deployed safely.

The core claim: **the tactical substrate is operationally distinct from cloud-native; the Edge Degraded Operations Undecalogy specifies the architecture that satisfies HGC³AE²'s commitments under tactical conditions.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Through pre-authorized envelopes that survive disconnection.

G — Curated Context. Local curated context survives disconnection; reconciliation merges with authoritative source on reconnection.

C¹ — Cybersecurity. Adversarial pressure assumed on every layer; security architecture is load-bearing in EDO-P2 and EDO-P9.

C² — Continuity. Continuity is the substrate's primary design property.

C³ — Coherence. Maintained across reconciliation cycles via CRDT/WAL primitives (EDO-P4).

A — Agentially Engineered. Within pre-authorized envelopes under disconnection.

E — Executed. Continuous execution under degraded conditions.

The exponent — the closure — is the disconnected-reconcile-adjust loop.

The remaining ten papers of the Edge Degraded Operations Undecalogy develop each architectural commitment in turn. EDO-P2 treats HGC³AE² at the degraded edge directly. EDO-P3 treats the DTN custody surface. EDO-P4 treats state coherence under partition. EDO-P5 treats out-of-path supervision. EDO-P6 treats protocol selection. EDO-P7 treats workload classification. EDO-P8–P10 treat classification, accreditation, and interoperability. EDO-P11 capstone integrates the architecture.¹⁹²⁰²¹²²

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19 EDO-P2 *HGC³AE² at the Degraded Edge* promotion to v1.0-preprint (in concurrent drafting; yks-build#104).

20 EDO-P3 *When the Link Is the Variable* promotion (in concurrent drafting; yks-build#95).

21 EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

22 EDO-P11 *Deployment Problem* CAPSTONE promotion (in concurrent drafting; yks-build#110).



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